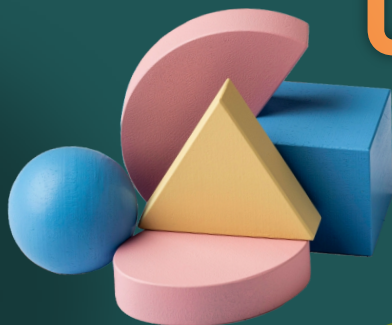


CO-ORDINATE GEOMETRY



Distance formula

$$P(x_1, y_1) \quad \bullet \text{-----} \bullet \quad Q(x_2, y_2)$$

Distance between two points $P(x_1, y_1)$ and $Q(x_2, y_2)$ is given by

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

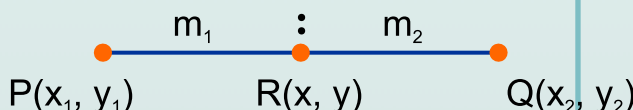


Section formula



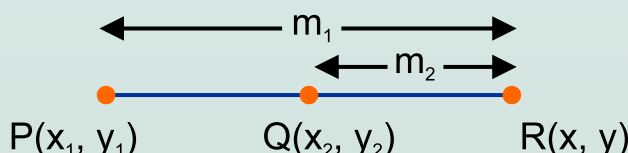
Formula for internal division : The coordinates of the point $R(x, y)$ which divides the join of two given points $P(x_1, y_1)$ and $Q(x_2, y_2)$ internally in the ratio $m_1 : m_2$ are given by

$$\left(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right)$$



Formula for external division : The coordinates of the point $R(x, y)$ which divides the join of two given points $P(x_1, y_1)$ and $Q(x_2, y_2)$ externally in the ratio $m_1 : m_2$ are given by

$$\left(\frac{m_1 x_2 - m_2 x_1}{m_1 - m_2}, \frac{m_1 y_2 - m_2 y_1}{m_1 - m_2} \right)$$



Mid-point formula : If R is the midpoint of PQ , then its coordinates are given by

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$





Centroid of a triangle



The point of concurrence of the medians of a triangle is called the centroid of triangle. It divides the median in the ratio 2:1.

The coordinates of the centroid of a triangle whose vertices are (x_1, y_1) , (x_2, y_2) and (x_3, y_3) are given by $\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$

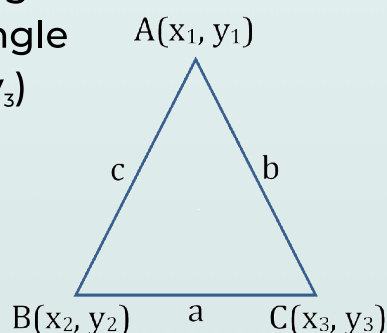
Incentre of a triangle



Incentre of a triangle is the point of concurrence of the internal bisectors of the angles of a triangle.

The coordinates of the incentre of a triangle whose vertices are (x_1, y_1) , (x_2, y_2) and (x_3, y_3) are given by

$$\left(\frac{ax_1 + bx_2 + cx_3}{a + b + c}, \frac{ay_1 + by_2 + cy_3}{a + b + c} \right)$$



Area of Triangle



The area of a triangle whose vertices are (x_1, y_1) , (x_2, y_2) and (x_3, y_3) is given by

$$A = \frac{1}{2} |[x_1 (y_2 - y_3) + x_2 (y_3 - y_1) + x_3 (y_1 - y_2)]|$$

Condition of collinearity of three points



The points $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ will be collinear (i.e., will lie on a straight line) if the area of the triangle assumed to be formed by joining them is zero.